安裝授權
ONTAP Systems Switches

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安裝適用於B3作業 叢集交換器的授權

B16-53248叢集交換器基礎模式已獲得16個10GbE或25GbE連接埠和2個100GbE連接埠的授權。您可以購買更多授權來新增連接埠。

下列授權可在B3交互 式叢集交換器上使用：

授權類型	授權詳細資料
支援的韌體版本	SW-BE-53248A1-G1-8P-LIC
Broadcom 8P 10-25、2P40至100授權金鑰、X190005/R	EFOS 3.4.3.3及更新版本
SW-BE-53248A1-G1-16P-LIC	Broadcom 16P 10-25、4P40-100授權金鑰、X190005/R
EFOS 3.4.3.3及更新版本	SW-BE-53248A1-G1-24P-LIC
Broadcom 24P 10-256P40-100授權金鑰、X190005/R	EFOS 3.4.3.3及更新版本
SW-BES54248-40-100G授權	Broadcom 6port 40G100G授權金鑰、X190005/R
EFOS 3.4.4.6及更新版本	SW-BES53248-8P-10G25G授權
Broadcom 8Port 10G25G授權金鑰、X190005/R	EFOS 3.4.4.6及更新版本
SW-BES53248-16P-1025G-LIC	Broadcom 16Port 10G25G授權金鑰、X190005/R
EFOS 3.4.4.6及更新版本	SW-BES53248-24P-1025G-LIC
Broadcom 24Port 10G25G授權金鑰、X190005/R	EFOS 3.4.4.6及更新版本

步驟

1. 將叢集交換器連接至管理網路。
2. 使用「ping」命令來驗證與裝載EFOS、授權和RCF檔案的伺服器之間的連線。

   此範例可驗證交換器是否連接至IP位址為172.19.2.1的伺服器：

```
(cs2) # ping 172.19.2.1
Pinging 172.19.2.1 with 0 bytes of data:

Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
```

3. 檢查交換器CS2目前的授權使用量：

「如何授權」

```
(cs2) # show license
Reboot needed..................................... No
Number of active licenses......................... 0

License Index  License Type     Status
-------------- ---------------- -----------

No license file found.
```

4. 安裝授權檔案。下列範例使用SFTP將授權檔案複製到金鑰索引1。

重複此步驟以載入更多授權、並使用不同的金鑰索引編號。

```
(cs2) #copy sftp://root@172.19.2.1/var/lib/tftpboot/license.dat
nvram:license-key 1
Remote Password:********

Mode............................................ SFTP
Set Server IP................................... 172.19.2.1
Path............................................ /var/lib/tftpboot/
Filename........................................ license.dat
Data Type....................................... license


Management access will be blocked for the duration of the transfer
Are you sure you want to start? (y/n) y

File transfer in progress. Management access will be blocked for the
duration of the transfer. Please wait...


License Key transfer operation completed successfully. System reboot is
required.
```

5. 顯示所有目前的授權資訊、並在重新啟動交換器CS2之前記下授權狀態：

「如何授權」

```
(cs2) # show license

Reboot needed..................................... Yes
Number of active licenses...................... 0


License Index  License Type      Status
-------------- ----------------- -------------------------------
1              Port              License valid but not applied
```

6. 顯示所有授權連接埠：

   「How port all | exclude Detach」

   交換器重新開機後、才會顯示其他授權檔的連接埠。

```
(cs2) # show port all \| exclude Detach

                 Admin     Physical   Physical   Link   Link    LACP
Actor
Intf      Type   Mode      Mode       Status     Status Trap    Mode
Timeout
--------- ------ --------- ---------- ---------- ------ ------- ------
--------
0/1              Disable   Auto                  Down   Enable  Enable
long
0/2              Disable   Auto                  Down   Enable  Enable
long
0/3              Disable   Auto                  Down   Enable  Enable
long
0/4              Disable   Auto                  Down   Enable  Enable
long
0/5              Disable   Auto                  Down   Enable  Enable
long
0/6              Disable   Auto                  Down   Enable  Enable
long
0/7              Disable   Auto                  Down   Enable  Enable
long
0/8              Disable   Auto                  Down   Enable  Enable
long
0/9              Disable   Auto                  Down   Enable  Enable
long
0/10             Disable   Auto                  Down   Enable  Enable
long
0/11             Disable   Auto                  Down   Enable  Enable
long
0/12             Disable   Auto                  Down   Enable  Enable
long
0/13             Disable   Auto                  Down   Enable  Enable
long
0/14             Disable   Auto                  Down   Enable  Enable
long
0/15             Disable   Auto                  Down   Enable  Enable
long
0/16             Disable   Auto                  Down   Enable  Enable
long
0/55             Disable   Auto                  Down   Enable  Enable
long
0/56             Disable   Auto                  Down   Enable  Enable
long
```

7. 重新啟動交換器：

   "重裝"

```
(cs2) # reload

The system has unsaved changes.
Would you like to save them now? (y/n) y

Config file 'startup-config' created successfully .

Configuration Saved!
Are you sure you would like to reset the system? (y/n) y
```

8. 請檢查新授權是否已啟用、並注意已套用授權：

   「如何授權」

```
(cs2) # show license

Reboot needed..................................... No
Number of installed licenses.................... 1
Total Downlink Ports enabled.................... 16
Total Uplink Ports enabled...................... 8

License Index  License Type              Status
-------------- -------------------------
------------------------------------
1              Port                      License applied
(cs2) #
```

9. 檢查所有新連接埠是否可用：

   「How port all | exclude Detach」

```
(cs2) # show port all \| exclude Detach

                 Admin     Physical   Physical   Link   Link    LACP
Actor
Intf      Type   Mode      Mode       Status     Status Trap    Mode
Timeout
--------- ------ --------- ---------- ---------- ------ ------- ------
--------
0/1              Disable    Auto                 Down   Enable  Enable
long
```

```
0/2              Disable    Auto                 Down   Enable  Enable
long
0/3              Disable    Auto                 Down   Enable  Enable
long
0/4              Disable    Auto                 Down   Enable  Enable
long
0/5              Disable    Auto                 Down   Enable  Enable
long
0/6              Disable    Auto                 Down   Enable  Enable
long
0/7              Disable    Auto                 Down   Enable  Enable
long
0/8              Disable    Auto                 Down   Enable  Enable
long
0/9              Disable    Auto                 Down   Enable  Enable
long
0/10             Disable    Auto                 Down   Enable  Enable
long
0/11             Disable    Auto                 Down   Enable  Enable
long
0/12             Disable    Auto                 Down   Enable  Enable
long
0/13             Disable    Auto                 Down   Enable  Enable
long
0/14             Disable    Auto                 Down   Enable  Enable
long
0/15             Disable    Auto                 Down   Enable  Enable
long
0/16             Disable    Auto                 Down   Enable  Enable
long
0/49             Disable   100G Full             Down   Enable  Enable
long
0/50             Disable   100G Full             Down   Enable  Enable
long
0/51             Disable   100G Full             Down   Enable  Enable
long
0/52             Disable   100G Full             Down   Enable  Enable
long
0/53             Disable   100G Full             Down   Enable  Enable
long
0/54             Disable   100G Full             Down   Enable  Enable
long
0/55             Disable   100G Full             Down   Enable  Enable
long
0/56             Disable   100G Full             Down   Enable  Enable
long
```

```
(cs2) #
```

限制與限制- Bes - 53248交換器

如果安裝授權時發生問題、請先執行下列偵錯命令、再重新執行「copy」命令來安裝授權。

要使用的除錯命令：

「Debuger傳輸除錯授權」

```
(cs2) # debug transfer
Debug transfer output is enabled.
(cs2) # debug license
Enabled capability licensing debugging.
```

當您在啟用「debug transfer」和「debug license」選項的情況下執行「copy」命令時、會傳回下列記錄輸出：

```
transfer.c(3083):Transfer process  key or certificate file type = 43
transfer.c(3229):Transfer process  key/certificate cmd = cp
/mnt/download//license.dat.1 /mnt/fastpath/ >/dev/null 2>&1CAPABILITY
LICENSING :
Fri Sep 11 13:41:32 2020: License file with index 1 added.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Validating hash value
29de5e9a8af3e510f1f16764a13e8273922d3537d3f13c9c3d445c72a180a2e6.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Parsing JSON buffer {
  "license": {
    "header": {
      "version": "1.0",
      "license-key": "964B-2D37-4E52-BA14",
      "serial-number": "QTFCU38290012",
      "model": "BES-53248"
  },
  "description": "",
  "ports": "0+6"
  }
}.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: License data does not
contain 'features' field.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Serial number
QTFCU38290012 matched.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Model BES-53248 matched.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Feature not found in
license file with index = 1.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Applying license file 1.
```

在偵錯輸出中檢查下列項目：

- 請檢查序號是否符合：「序號QTFC38290012相符」
- 檢查交換器機型是否符合「model Bes - 53248 matched」
- 檢查之前是否未使用指定的授權索引。如果已使用授權索引、則會傳回下列錯誤：「License file /mnt/download//license．dat.1 already exists（授權檔案/mnt/download//license．dat.1已存在）」
- 連接埠授權並非功能授權。因此、預期會有下列陳述：「在索引= 1的授權檔案中找不到功能。」

使用「copy」命令將連接埠授權備份到伺服器：

```
(cs2) # copy nvram:license-key 1
scp://<UserName>@<IP_address>/saved_license_1.dat
```

請參閱 "安裝B封閉 環境叢集交換器的授權" 以取得可用授權所支援的韌體版本詳細資料。

如果您需要將交換器軟體從3.4.4.6版降級、則會移除授權。這是預期的行為。

您必須先安裝適當的舊版授權、才能還原至舊版軟體。

編輯參考組態檔案（RCF）

若要啟動新授權的連接埠、您必須編輯最新版的RCF、然後取消註釋適用的連接埠詳細資料。預設授權會啟動連接埠0/1到0/16和0/55到0/56、而新授權的連接埠則會在連接埠0/17到0/54之間、視可用授權的類型和數量而定。

如果您嘗試編輯先前安裝的RCF、程序可能會失敗、因為RCF中有其他區域的現有組態、請參閱 [Edit a previously installed RCF file]。

如需有關用於B3作業 叢集交換器的可用授權類型詳細資料、請參閱 "安裝B封閉 環境叢集交換器的授權"。

例如、若要啟動SW-BES54248-40-100g-LIC授權、您必須取消註解RCF中的下列章節：

```
.
.
!
! 2-port or 6-port 40/100GbE node port license block
!
interface 0/49
no shutdown
description "40/100GbE Node Port"
!speed 100G full-duplex
speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/50
no shutdown
description "40/100GbE Node Port"
!speed 100G full-duplex
speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
```

```
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/51
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/52
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/53
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
```

```
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/54
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
.
.
```

> 對於0/49到0/54（含）之間的高速連接埠、請針對每個連接埠取消註解、但在RCF中只取消註釋一條*速度*線路、以便針對每個連接埠：

- 速度100公克全雙工
- 速度40G全雙工

如範例所示。

對於0/17到0/48（含）之間的低速度連接埠、請在啟用適當授權後、取消註釋整個8埠區段。

編輯先前安裝的RCF檔案

編輯之前安裝的RCF檔案並執行「套用指令碼」之後、您可能會收到下列錯誤訊息：

```
(CS1) #script apply BES-53248_RCF_v1.6-Cluster-HA.scr
 Are you sure you want to apply the configuration script? (y/n)
```

選取「y」之後、您會收到下列錯誤訊息：

```
config
 ...
 match cos 5
 Unrecognized command : match cos 5
 Error! in configuration script file at line number 40.
 CLI Command :: match cos 5.
 Aborting script.
```

若要避免或解決此問題、您可以選擇下列其中一個選項：

- 若要避免錯誤、您可以使用下列程序：
  a. 建立第二個僅包含新連接埠組態的RCF檔案。
  b. 將第二個RCF檔案複製到交換器。
  c. 將指令碼套用至交換器：

     適用記錄
- 若要解決此錯誤、請參閱知識庫文章： "錯誤！套用新的RCF時、設定指令碼檔案中的行號XX"

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